CARBON MARKETS CCUS AND DIRECT AIR CAPTURE

CARBON-UNIT IDENTITY FORK → MARKET ARCHITECTURE MATRIX → CLAIM-TYPE LADDER → INTEGRITY GATE → CREDIT LIFECYCLE RAIL → ACCOUNTING-RISK FIREWALL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g5 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 CARBON-UNIT IDENTITY FORK

CARBON-UNIT IDENTITY FORK | PERMISSION UNIT INSIDE A COMPLIANCE CONSTRAINT | VERIFIED OUTCOME AGAINST AN APPROVED BASELINE | ALLOCATION OR AUCTION VERSUS METHODOLOGY AND ISSUANCE | SURRENDER OR RETIREMENT UNDER THE APPLICABLE RULE | TRADABILITY DOES NOT ERASE DIFFERENT UNIT ORIGINS | MUST REMEMBER | CARBON PRICING AND MARKETS ASSIGN TRADABLE OR FISCAL INCENTIVES

ALLOWANCE → permission unit inside a compliance constraint

CREDIT → verified outcome against an approved baseline → ORIGIN → allocation or auction versus methodology and issuance → USE → surrender or retirement under the applicable rule

CORE CLAIM
- ALLOWANCE → permission unit inside a compliance constraint
- CREDIT → verified outcome against an approved baseline

MECHANISM
- ORIGIN → allocation or auction versus methodology and issuance
- USE → surrender or retirement under the applicable rule

CONSEQUENCE / LIMIT
- TRAP → tradability does not erase different unit origins
- MUST REMEMBER: Carbon pricing and markets assign tradable or fiscal incentives to...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → tradability does not erase different unit origins.

01

STAGE 01 MARKET ARCHITECTURE MATRIX

MARKET ARCHITECTURE MATRIX | CAP AND TRADE | ABSOLUTE COVERED-EMISSIONS CEILING | BASELINE AND CREDIT | PERFORMANCE AGAINST TARGET OR BASELINE | LEGAL OBLIGATION AND SURRENDER REQUIREMENT | NON-MANDATED PURCHASE OR CLAIM | WHAT CONSTRAINS EMISSIONS AND WHO MUST ACT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CAP AND TRADE → absolute covered-emissions ceiling	COMPLIANCE → legal obligation and surrender requirement	QUESTION → what constrains emissions and who must act
BASELINE AND CREDIT → performance against target or baseline	VOLUNTARY → non-mandated purchase or claim	CAP AND TRADE → absolute covered-emissions ceiling

CORE CLAIM
- CAP AND TRADE → absolute covered-emissions ceiling
- BASELINE AND CREDIT → performance against target or baseline

MECHANISM
- COMPLIANCE → legal obligation and surrender requirement
- VOLUNTARY → non-mandated purchase or claim

CONSEQUENCE / LIMIT
- QUESTION → what constrains emissions and who must act
- CAP AND TRADE → absolute covered-emissions ceiling

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUESTION → what constrains emissions and who must act.

02

STAGE 02 CLAIM-TYPE LADDER

CLAIM-TYPE LADDER | COUNTERFACTUAL SOURCE DOES NOT ARISE | EXISTING SOURCE EMITS LESS | CARBON LEAVES AMBIENT ATMOSPHERE | DURATION AND REVERSAL RISK DETERMINE DURABILITY | NO MERGER | EACH SUPPORTS A DIFFERENT CLIMATE CLAIM

AVOIDANCE → counterfactual source does not arise

REDUCTION → existing source emits less → REMOVAL → carbon leaves ambient atmosphere → STORAGE → duration and reversal risk determine durability

CORE CLAIM
- AVOIDANCE → counterfactual source does not arise
- REDUCTION → existing source emits less

MECHANISM
- REMOVAL → carbon leaves ambient atmosphere
- STORAGE → duration and reversal risk determine durability

CONSEQUENCE / LIMIT
- NO MERGER → each supports a different climate claim
- AVOIDANCE → counterfactual source does not arise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO MERGER → each supports a different climate claim.

03

STAGE 03 INTEGRITY GATE

INTEGRITY GATE | DEFINE THE COUNTERFACTUAL | TEST BUSINESS-AS-USUAL RISK | QUANTIFY BOUNDARY PERIOD AND OUTCOME | INDEPENDENT EVIDENCE CHECK | UNIT CREATED ONLY AFTER RULE-DEFINED GATES

REDUCTION → existing source emits less → REMOVAL → carbon leaves ambient atmosphere → STORAGE → duration and reversal risk determine durability

CORE CLAIM
- AVOIDANCE → counterfactual source does not arise
- REDUCTION → existing source emits less

MECHANISM
- REMOVAL → carbon leaves ambient atmosphere
- STORAGE → duration and reversal risk determine durability

CONSEQUENCE / LIMIT
- NO MERGER → each supports a different climate claim
- AVOIDANCE → counterfactual source does not arise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO MERGER → each supports a different climate claim.

03

STAGE 03 INTEGRITY GATE

INTEGRITY GATE · DEFINE THE COUNTERFACTUAL · TEST BUSINESS-AS-USUAL RISK · QUANTIFY BOUNDARY PERIOD AND OUTCOME · INDEPENDENT EVIDENCE CHECK · UNIT CREATED ONLY AFTER RULE-DEFINED GATES

BASELINE → define the counterfactual → ADDITIONALITY → test business-as-usual risk → MRV → quantify boundary period and outcome → VERIFICATION → independent evidence check → ISSUANCE → unit created only after rule-defined gates

CORE CLAIM
- BASELINE → define the counterfactual
- ADDITIONALITY → test business-as-usual risk

MECHANISM
- MRV → quantify boundary period and outcome
- VERIFICATION → independent evidence check

CONSEQUENCE / LIMIT
- ISSUANCE → unit created only after rule-defined gates
- BASELINE → define the counterfactual

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ISSUANCE → unit created only after rule-defined gates.

04

STAGE 04 CREDIT LIFECYCLE RAIL

CREDIT LIFECYCLE RAIL · DESIGN OR VALIDATION · METHOD AND PROJECT ACCEPTED · ACTIVITY AND OUTCOME RECORDED · EVIDENCE EXAMINED · ISSUANCE AND TRANSFER · TRADABLE UNIT ENTERS REGISTRY · UNIT REMOVED FROM CIRCULATION FOR A CLAIM

DESIGN OR VALIDATION → method and project accepted → MONITORING → activity and outcome recorded → VERIFICATION → evidence examined → ISSUANCE AND TRANSFER → tradable unit enters registry → RETIREMENT → unit removed from circulation for a claim

CORE CLAIM
- DESIGN OR VALIDATION → method and project accepted
- MONITORING → activity and outcome recorded

MECHANISM
- VERIFICATION → evidence examined
- ISSUANCE AND TRANSFER → tradable unit enters registry

CONSEQUENCE / LIMIT
- RETIREMENT → unit removed from circulation for a claim
- DESIGN OR VALIDATION → method and project accepted

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RETIREMENT → unit removed from circulation for a claim.

05

STAGE 05 ACCOUNTING-RISK FIREWALL

ACCOUNTING-RISK FIREWALL · DOUBLE ISSUANCE · MORE THAN ONE UNIT FOR ONE OUTCOME · DOUBLE USE · ONE UNIT USED MORE THAN ONCE · DOUBLE CLAIMING · SAME OUTCOME CLAIMED BY MULTIPLE ACTORS · CORRESPONDING ADJUSTMENT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
DOUBLE ISSUANCE → more than one unit for one outcome	DOUBLE CLAIMING → same outcome claimed by multiple actors	RULE → adjustment is not issuance transfer or retirement
DOUBLE USE → one unit used more than once	CORRESPONDING ADJUSTMENT → international accounting treatment	DOUBLE ISSUANCE → more than one unit for one outcome

CORE CLAIM
- DOUBLE ISSUANCE → more than one unit for one outcome
- DOUBLE USE → one unit used more than once

MECHANISM
- DOUBLE CLAIMING → same outcome claimed by multiple actors
- CORRESPONDING ADJUSTMENT → international accounting treatment

CONSEQUENCE / LIMIT
- RULE → adjustment is not issuance transfer or retirement
- DOUBLE ISSUANCE → more than one unit for one outcome

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → adjustment is not issuance transfer or retirement.

06

STAGE 06 INDIA CCTS STAGE LADDER

INDIA CCTS STAGE LADDER · LEGAL BASIS · ENABLING FRAMEWORK · SCHEME DESIGN · COMPLIANCE AND OFFSET PATHWAYS · TARGET OR METHODOLOGY · OBLIGATION AND CREDIT RULES · ISSUANCE AND TRADING

LEGAL BASIS → enabling framework

SCHEME DESIGN → compliance and offset pathways → TARGET OR METHODOLOGY → obligation and credit rules → ISSUANCE AND TRADING → operational transactions

CORE CLAIM
- LEGAL BASIS → enabling framework

MECHANISM
- TARGET OR METHODOLOGY → obligation and credit rules

CONSEQUENCE / LIMIT
- PRICE AND VOLUME → dated market evidence only

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → adjustment is not issuance transfer or retirement.

06

STAGE 06

INDIA CCTS STAGE LADDER

INDIA CCTS STAGE LADDER | LEGAL BASIS | ENABLING FRAMEWORK | SCHEME DESIGN | COMPLIANCE AND OFFSET PATHWAYS | TARGET OR METHODOLOGY | OBLIGATION AND CREDIT RULES | ISSUANCE AND TRADING

LEGAL BASIS → enabling framework

SCHEME DESIGN → compliance and offset pathways → TARGET OR METHODOLOGY → obligation and credit rules → ISSUANCE AND TRADING → operational transactions

CORE CLAIM
- LEGAL BASIS → enabling framework
- SCHEME DESIGN → compliance and offset pathways

MECHANISM
- TARGET OR METHODOLOGY → obligation and credit rules
- ISSUANCE AND TRADING → operational transactions

CONSEQUENCE / LIMIT
- PRICE AND VOLUME → dated market evidence only
- LEGAL BASIS → enabling framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRICE AND VOLUME → dated market evidence only.

07

STAGE 07

CCUS CHAIN

CCUS CHAIN | CONCENTRATED CARBON DIOXIDE STREAM | CAPTURE AND CONDITIONING | SEPARATION AND PREPARATION | PIPELINE SHIP ROAD OR OTHER ROUTE | UTILISATION OR STORAGE | DESTINATION DETERMINES DURABILITY | MONITORING AND CLOSURE

SOURCE → concentrated carbon dioxide stream → CAPTURE AND CONDITIONING → separation and preparation → TRANSPORT → pipeline ship road or other route → UTILISATION OR STORAGE → destination determines durability → MONITORING AND CLOSURE → permanence and responsibility

CORE CLAIM
- SOURCE → concentrated carbon dioxide stream
- CAPTURE AND CONDITIONING → separation and preparation

MECHANISM
- TRANSPORT → pipeline ship road or other route
- UTILISATION OR STORAGE → destination determines durability

CONSEQUENCE / LIMIT
- MONITORING AND CLOSURE → permanence and responsibility
- SOURCE → concentrated carbon dioxide stream

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSPORT → pipeline ship road or other route.

08

STAGE 08

UTILISATION-STORAGE TEST

UTILISATION-STORAGE TEST | PRODUCT USE | HOW LONG IS CARBON RETAINED | PROCESS USE | DOES LATER RELEASE OCCUR | GEOLOGICAL STORAGE | CONTAINMENT AND MONITORING | REVERSAL RISK AND REMEDIATION

PRODUCT USE → how long is carbon retained

PROCESS USE → does later release occur → GEOLOGICAL STORAGE → containment and monitoring → LEAKAGE → reversal risk and remediation

CORE CLAIM
- PRODUCT USE → how long is carbon retained
- PROCESS USE → does later release occur

MECHANISM
- GEOLOGICAL STORAGE → containment and monitoring
- LEAKAGE → reversal risk and remediation

CONSEQUENCE / LIMIT
- NET BENEFIT → full lifecycle accounting required
- PRODUCT USE → how long is carbon retained

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRODUCT USE → how long is carbon retained.

09

STAGE 09

CAPTURE TECHNOLOGY COMPARISON

CAPTURE TECHNOLOGY COMPARISON | POINT SOURCE CCUS | CARBON INTERCEPTED BEFORE RELEASE | CARBON SEPARATED FROM AMBIENT AIR | BIOGENIC CARBON PLUS CAPTURE AND STORAGE | REMOVAL TEST | ATMOSPHERIC ORIGIN PLUS DURABLE STORAGE | LIFECYCLE TEST

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
POINT SOURCE CCUS → carbon intercepted before release	BECCS → biogenic carbon plus capture and storage	LIFECYCLE TEST → energy land transport and reversal
DAC → carbon separated from ambient air	REMOVAL TEST → atmospheric origin plus durable storage	CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not...

CORE CLAIM
- POINT SOURCE CCUS → carbon intercepted before release
- DAC → carbon separated from ambient air

MECHANISM
- BECCS → biogenic carbon plus capture and storage
- REMOVAL TEST → atmospheric origin plus durable storage

CONSEQUENCE / LIMIT
- LIFECYCLE TEST → energy land transport and reversal
- CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not.

09

STAGE 09 CAPTURE TECHNOLOGY COMPARISON

CAPTURE TECHNOLOGY COMPARISON · POINT SOURCE CCUS · CARBON INTERCEPTED BEFORE RELEASE · CARBON SEPARATED FROM AMBIENT AIR · BIOGENIC CARBON PLUS CAPTURE AND STORAGE · REMOVAL TEST · ATMOSPHERIC ORIGIN PLUS DURABLE STORAGE · LIFECYCLE TEST

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
POINT SOURCE CCUS → carbon intercepted before release	BECCS → biogenic carbon plus capture and storage	LIFECYCLE TEST → energy land transport and reversal
DAC → carbon separated from ambient air	REMOVAL TEST → atmospheric origin plus durable storage	CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not...

CORE CLAIM

- POINT SOURCE CCUS → carbon intercepted before release
- DAC → carbon separated from ambient air

MECHANISM

- BECCS → biogenic carbon plus capture and storage
- REMOVAL TEST → atmospheric origin plus durable storage

CONSEQUENCE / LIMIT

- LIFECYCLE TEST → energy land transport and reversal
- CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not.

10

STAGE 10 NET-ZERO MITIGATION HIERARCHY

NET-ZERO MITIGATION HIERARCHY · PREVENT UNNECESSARY EMISSIONS · DECARBONISE EXISTING ACTIVITIES · ADDRESS HARD-TO-ABATE POINT SOURCES · COUNTERBALANCE GENUINELY RESIDUAL EMISSIONS · NET BALANCE NEEDS ONE BOUNDARY AND PERIOD · EVIDENCE LIMIT · FIX SYSTEM BOUNDARY, BASELINE, UNIT

AVOID → prevent unnecessary emissions

REDUCE → decarbonise existing activities → CAPTURE → address hard-to-abate point sources → REMOVE → counterbalance genuinely residual emissions

CORE CLAIM

- AVOID → prevent unnecessary emissions
- REDUCE → decarbonise existing activities

MECHANISM

- CAPTURE → address hard-to-abate point sources
- REMOVE → counterbalance genuinely residual emissions

CONSEQUENCE / LIMIT

- VERIFY → net balance needs one boundary and period
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit,.

11

STAGE 11 CARBON INSTRUMENTS ANSWER SPINE

CARBON INSTRUMENTS ANSWER SPINE · UNIT MARKET AND CLAIM TYPE · BASELINE MRV ISSUANCE TRANSFER RETIREMENT · CAPTURE TRANSPORT UTILISATION OR STORAGE · PERMANENCE LEAKAGE LIFECYCLE AND DOUBLE COUNTING · CURRENT RULE PRICE VOLUME COST AND CAPACITY SOURCE

CLASSIFY → unit market and claim type → TRACE → baseline MRV issuance transfer retirement → FOLLOW → capture transport utilisation or storage → TEST → permanence leakage lifecycle and double counting → AUDIT → current rule price volume cost and capacity source

CORE CLAIM

- CLASSIFY → unit market and claim type
- TRACE → baseline MRV issuance transfer retirement

MECHANISM

- FOLLOW → capture transport utilisation or storage
- TEST → permanence leakage lifecycle and double counting

CONSEQUENCE / LIMIT

- AUDIT → current rule price volume cost and capacity source
- CLASSIFY → unit market and claim type

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AUDIT → current rule price volume cost and capacity source.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · DAC'S FUNDAMENTAL TECHNICAL CHALLENGE IS THERMODYNAMIC · ATMOSPHERIC CO_2 CONCENTRATION · POLICY IMPLICATION · BECAUSE DAC DELIVERS GENUINE ATMOSPHERIC REMOVAL (UNLIKE · GLOBAL CCUS AND DAC DEPLOYMENT CAPACITY REMAINS SMALL RELATIVE · THE INDIAN CARBON MARKET'S SPECIFIC SECTORAL COVERAGE, TARGET-SETTING METHODOLOGY · MRV-SYSTEM ROBUSTNESS FOR BOTH INDIA'S DOMESTIC CARBON MARKET AND

OPTIONAL SOURCE DEPTH

- CAUTION: DAC's fundamental technical challenge is thermodynamic: atmospheric CO_2 concentration
- CAUTION: Policy implication: because DAC delivers genuine atmospheric removal (unlike
- CAUTION: Global CCUS and DAC deployment capacity remains small relative to the scale IPCC
- CAUTION: The Indian Carbon Market's specific sectoral coverage, target-setting methodology and

ADVANCED DEBATE

- CAUTION: MRV-system robustness for both India's domestic carbon market and international
- India's carbon market has a two-track design: a compliance mechanism (building on PAT,
- Standard fossil-source CCUS is an emissions-avoidance technology; genuine atmospheric
- DAC's higher cost/energy-intensity relative to CCUS stems from the much lower CO_2

LEGACY / NUANCE

- Net-zero frameworks require genuine removal technologies (DAC/BECCS/nature-based sinks)
- NOT: All CCUS applications, including standard fossil-power-plant CCUS, deliver negative
- NOT: India's carbon market is a single-track, purely compliance-based system. - It has a
- NOT: DAC is currently deployed at a scale comparable to global CCUS capacity. - DAC remains

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels